Know-your-own-League context: insights for player preparation and recruitment - Part 4: match-to-match player rotations

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Soccer | Association football | Team performance | Home and away | Line-up changes

Headline

Due to the potential tradeoff between line-up stability for optimal match performance (i.e., which favours better between-player communication, 13) and load management (i.e., to preserve players' health, especially when playing congested), deciding how many players to rotate from one match to the next, and who to rotate is a key question for every manager (10). Following our work on team formations (5), results and goals scored (6), and player substitutions (7), this fourth part of our series of short papers focuses on player management, i.e., match-to-match player rotations over the past two decades in the top seven European leagues.

Aim

This study first aims to examine the between-league differences in player match-to-match rotations between matches. These were all examined for both home vs. away matches. The second aim is to identify the key factors that could affect coaches' rotation strategies using machine learning models, i.e., Extreme Gradient Boosting (XGBoost) for Poisson regression.

Methods

Data extraction

We extracted the fixtures data of teams competing in the top seven European leagues, i.e., the English Premier League (EPL), the French Ligue 1, the German Bundesliga, the Dutch Eredivisie, Italian Serie A, the Portuguese Liga and Spanish Liga. This includes 21 seasons from 2001/02 to 2021/22. The fixtures from all in-season competitions were considered, i.e., leagues and domestic, European and international cups. This data was sourced from Transfermarkt. Across the 21 seasons, this represents 43 competitions, 269 teams and more than 61,000 fixtures.

Descriptive Analysis

For match-to-match rotations, we computed the number of match starters who did not start the previous match. All in-season competitions were included in this assessment: domestic league and cups, and European/international cups, but only the observations from domestic league matches were reported. For instance, let's assume Team A plays a league match (L1) on Sunday, seven days after a domestic cup match (C), then a Champions League match (CL) on Wednesday and finally another league game on Saturday (L2). The C and CL matches are used to calculate the match-to-match rotations of L1 and L2, respectively, but only L1 and L2 are reported in the analysis. To display match-to-match player rotations, we used boxplots to convey the spread across teams within the league. Boxplots display the median, with the size of the box (Interquartile range, IQR) spanning from Q1 (25th percentile) to Q3 (75th percentile). The outside bars are the lower and upper whiskers: the lower whisker is the smallest value in the dataset that is no smaller than Q1 - 1.5 x IQR and the upper whisker is the largest value that is no greater than Q3 + 1.5 x IQR. Values outside of both whiskers are considered outliers and are not shown here. The observations used to create those boxplots are team averages per season. For a given league and season, the box is then based on N means - N being the number of teams within the given league during the given season.

Model-based Analysis

Based on the above-mentioned fixtures data extract, the following metrics were used to identify the key factors that could affect coaches' rotation strategies:

- Rest days: number of days between matches, irrespective of the competition. For instance, playing on Wednesday after a previous game on Sunday gives two rest days (Monday and Tuesday).
- Previous game travel distance: Haversine distance in km, i.e., the angular distance between two points on the surface of a sphere.
- Team's Elo rating: its definition is based on the standard Elo rating formula and our own assumptions (see Appendix A).
- Difference between teams Elo rating and the opponents' rating.
- Recent performance: percentage of points won during the last five games from all competitions. A cup win is considered as 3 points, a loss 0 point.
- Previous competition split into four categories:
  - Domestic league
  - Domestic cups
  - Major international cups: Champions League & Europa League (qualifiers included), UEFA Super Cup and FIFA Club World Cup
  - Minor international cups: UEFA Intertoto Cup (until 2009) and Europa Conference League (qualifiers included)
- Home or away

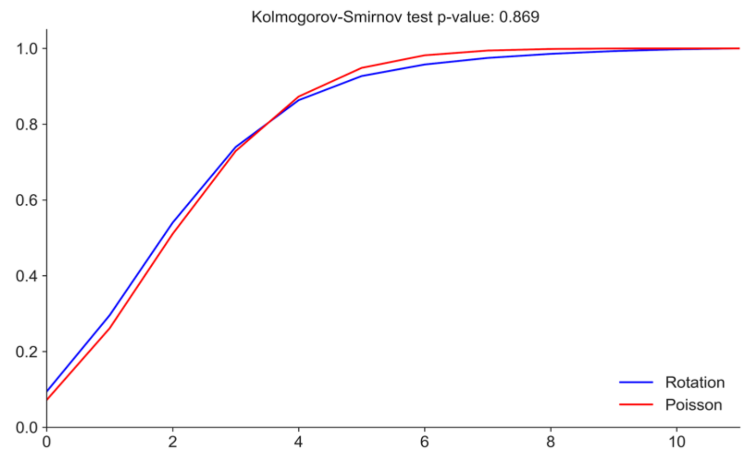

Fig. 1. Cumulative distribution function (CDF) of the Rotation variable (blue) and a Poisson variable (red) whose mean is equal to the Rotations mean in the data set. The one-sample Kolmogorov-Smirnov test is used to assess the similarity between the 2 CDFs.

The one-sample Kolmogorov-Smirnov (KS) test was used to compare the cumulative distribution function (CDF) of the Rotation response variable and reference probability distributions. The most similar CDF was the one from a Poisson variable whose mean was equal to the Rotations mean in the data set, i.e., 2.63 rotations (Figure 1). The high p-value (0.869) of the KS test confirmed that we can assume the response variable is drawn from the following Poisson distribution.

In order to study the relationship between the rotation response variable and the explanatory variables, we used eXtreme Gradient Boosting (XGBoost) for Poisson regression. This can be challenging with real-life data as it assumes equidispersion, i.e., the mean of the distribution is equal to the variance. In our case, this is not entirely true as the relative variance - which is the ratio of the variance to the mean - is equal to 1.37. However, as the over-dispersion is inferior to two, it does not significantly impact our model (1). Even if multicollinearity has no impact on performance of XGBoost models by nature, it does affect feature importance. Thereby, we studied the Pearson correlation between all pairs of the 10 features. The largest positive correlation value was between Elo and Elo Difference (0.66), while the largest negative one was between League and Domestic Cup (-0.69). Considering that there was no very strong (positive or negative) correlation between any pairs of features, we decided to keep all of them in the model. The data set is also almost complete as the only two variables containing missing data are rest days (0.01%) and previous game travel distance (1.67%) with low proportion.

As teams played in empty stadiums from COVID restart in May 2020 to the end of the 2020/21 season—impacting the home-field advantage—we decided to exclude this period from the analysis. We then split the data set into training and test sets based on the following seasons:

- Training set (80% of the data): from 2001/02 to 2016/17
- Test set (remaining 20%): from 2017/18 to 2021/22 excluding the COVID period

The XGBoost model structure is defined by hyperparameters. In order to get the optimal ones, we used a randomised grid-search based on the training set. A grid-search runs through all the different parameters fed into the parameter grid and produces their best combination based on a scoring metric, i.e., Poisson deviance here. A randomised grid-search does not try all combinations out but rather a fixed number of iterations, i.e., 20 here. It then returns a relatively accurate performing model in a significantly shorter runtime (Bergstra & Bengio, 2012). The parameter grid considered is the following one:

- Number of iterations: 20
- Model evaluation: Poisson deviance
- Cross-validation (CV) splitting strategy: 5-fold CV
- Step size shrinkage: 0.1*, 0.15, 0.2
- Subsample ratio of columns when constructing each tree: 0.3, 0.5*, 0.7
- Maximum depth of a tree: 3, 7*, 15
- Number of trees: 20, 100 500*
- L1 regularisation term on weights: 0.1, 10, 50*

The values with an asterisk are the optimal ones for the model.

To explain how our model works, i.e., the decisions it is making, we used SHapley Additive exPlanations (SHAP) values (3). One of the fundamental properties of SHAP values is that all the input features will always sum up to the difference between baseline (expected) model output and the current model output for the prediction being explained. Considering our topic, we decided to study the following concepts:

- Feature Importance: based on the absolute SHAP values per feature, it analyses the global importance of each feature.
- Feature Relationship: it indicates the relationship between the value of a feature and the impact on the prediction.
- Feature Dependence: as an alternative to partial dependence plots and accumulated local effects, it focuses on a given feature and shows for each data instance the relationship between the feature value (x-axis) and the corresponding SHAP value (y-axis).

Based on this Feature Dependence concept, we created our own version called SHAP dependency plot (Figures 8 to 13). When the feature (x-axis) is non-binary, local trends between two non-binary entities are shown using locally-linear estimated scatterplot smoothing (LOESS) and its 95% prediction intervals (PI). We defined four zones to summarise the impact of the feature on match-to-match rotations:

- Increasing: both the LOESS curve and PI are above 0

- ◦ Likely increasing: the LOESS curve is above 0 but 0 is included in PI
- ◦ Likely decreasing: the LOESS curve is below 0 but 0 is included in PI
- ◦ Decreasing: both the LOESS curve and PI are below 0

For binary features, a box plot is used. It is based on the SHAP values related to each instance (0 and 1), and the same parameters as the ones defined in the above-mentioned Descriptive Analysis subsection. The four zones are created as follows:

- ◦ Increasing: both the median and interval between whiskers (WI) are above 0
- ◦ Likely increasing: the median is above 0 but 0 is included in the WI
- ◦ Likely decreasing: the median is below 0 but 0 is included in the WI
- ◦ Decreasing: both the median curve and WI are below 0

Finally, we evaluated the suitability of our model with the Poisson Deviance on the test set. We compared our model trained on all listed features with:

- Naive model: all predictions are the rotation mean in the training set
- Rest days model: all predictions are calculated from a model only trained on the rest days variable. We chose rest days as it is the most important feature for our model. This then enabled us to study the significance of the impact of the other variables.

Knowing that the model performance can vary depending on the test set, we generated 95% confidence intervals (CI) for the Poisson deviance by bootstrapping the test set. The full model performs better than both the rest days and naive models as seen in Figure 2.

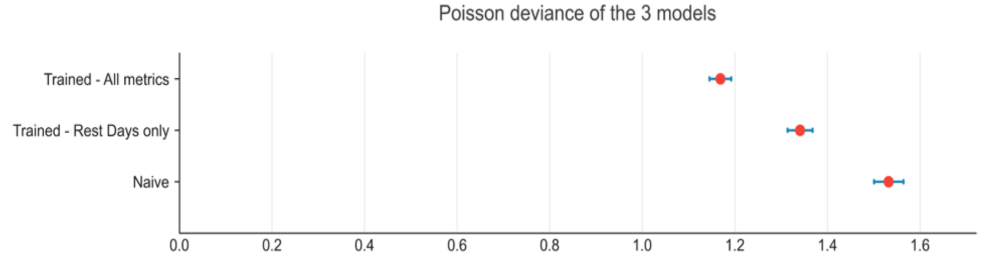

Fig. 2. Poisson deviance of the model trained on all metrics, the one trained only on rest days variable and the naive one.

Discussion and Results

This study is the first to our knowledge to examine between-league differences in player management, namely match-to-match player rotations, and their determinants via modelling, i.e., XGBoost for Poisson Regression. Since there is likely a tradeoff between line-up stability for optimal match performance (10, 13) and load management to preserve players' health, deciding how many players to rotate from one match to the next, and who to rotate is a key question for every manager.

Descriptive analysis

The team-average number of match-to-match line-up rotations (home and away matches) per season over the past two decades is shown for each league individually in Figures 3 and 4. Overall, the medians of team-average match-to-match rotations were between two and three, with coaches of Ligue 1 and Eredivisie tending to perform the most (around three) and the least (less than two) number of rotations, respectively. While the respective importance of some factors that may explain rotations are discussed below (modelling part), the greater team formations variability in Ligue 1 vs. Eredivisie (5) may also partly explain those between-league differences in match-to-match rotations. Indeed, players rotations are often required to support formation changes (e.g., a third central defender on, and one offensive middlefield out when switching from a 4-3-3 to 3-5-1). Overall, present results show that rotations have remained stable across the years and there doesn't seem to be an effect of match location either, with coaches rotating about three players per match. The relative stability over time in terms of player turnover strategies is consistent with a recent report from the CIES Football Observatory (12). Our findings suggest a generic (universal) approach among coaches when it comes to managing players, where they need to balance player welfare and team performance (with the lower number of rotations, the better the between-player communication) (13).

When looking closer at the distribution of match-to-match rotations (Figures 3 and 4), there were some small increasing trends across the years in some leagues (i.e., from two to three in the Premier League and La Liga). There were also between-league differences in the spread of data between teams within the same league, e.g., larger whiskers and percentiles within the EPL vs. the Eredivisie. Altogether, this suggests less homogeneity in rotation strategies in the EPL than in the Eredivisie. In the EPL there are likely larger differences in team wealth and strength between the top and bottom teams (e.g., Liverpool and Man City vs. Southampton or Nottingham Forest); top teams presumably have a better squad and better bench players so that they can likely rotate more players from a match to the next and still perform well - which lower ranked teams can't often afford (Buchheit 2022). Best teams in top leagues may also have to play both more matches in their domestic league (i.e., the Premier League is composed of 20 teams, while Eredivisie, of 18 only) and more matches of higher relative importance (more teams are involved in European competitions in Premier League than in Eredivisie), which likely also push them to rotate more players.

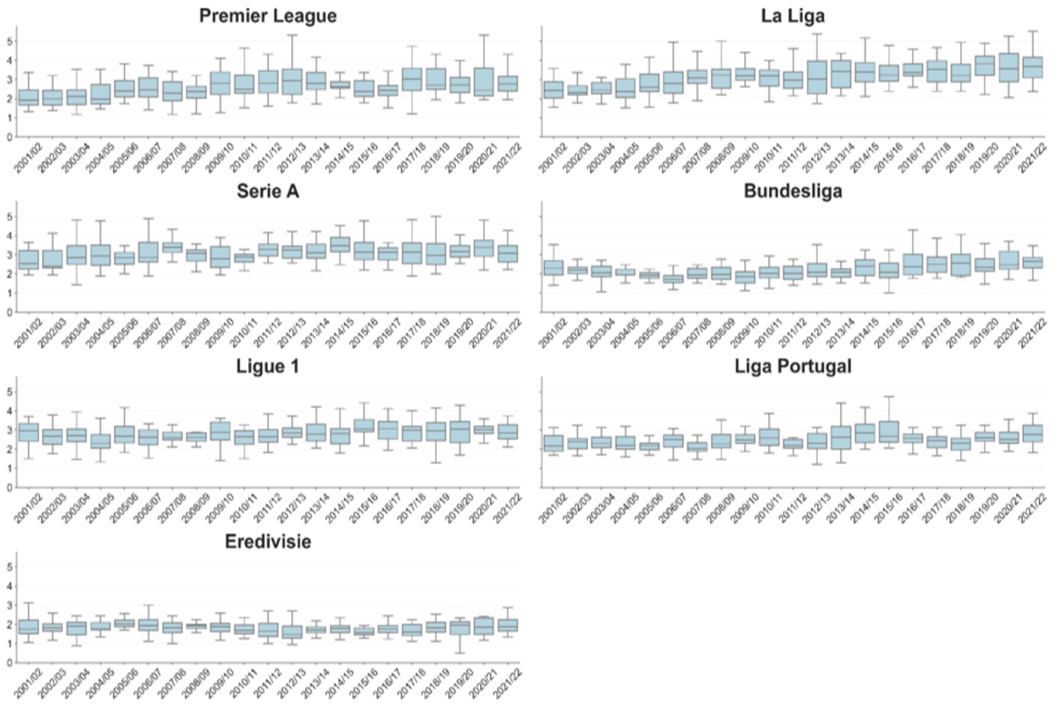

Fig. 3. Team-average number of home match-to-match line-up rotations per season in all leagues over the past two decades.

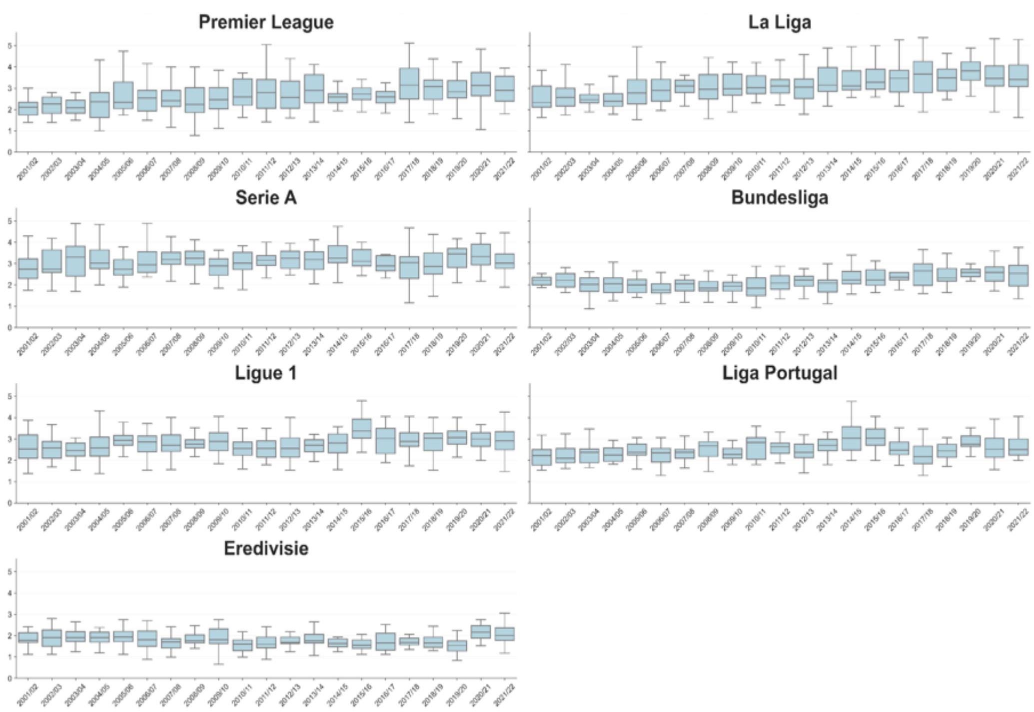

Fig. 4. Team-average number of away match-to-match line-up rotations per season in all leagues over the past two decades.

Model-based Analysis

This is to our knowledge the first time that rotation strategies have been modelled, here using XGBoost for Poisson Regression. We use SHAP to interpret the model predictions. Its additive nature allows each prediction to be split out into the contributions from each feature in the model over and above the SHAP expected value, and ranked in order of importance/magnitude. The SHAP expected value is essentially the average number of rotations observed across the entire dataset, 2.671 in our case. Figure 5 shows how SHAP is used to interpret the model prediction for the game between Real Madrid and Cadiz on May 15, 2022 after Real Madrid had played against Levante on May 12, 2022. The combined SHAP contributions plus the expected value gives the resulting prediction, 4.336 rotations in this case. The main contributor here, Real Madrid's Elo, was 1773 before the game and this contributed 1.13 rotations in addition to the average value. The contribution of the other features is shown in Figure 5. Real Madrid actually rotated six players against Cadiz.

Looking across all model predictions, the SHAP contributions for all metrics in the model can be used to provide insight into the key factors affecting coaches' rotation strategies in general. In Figure 6 these are ranked in relation to their respective overall contribution to the model, either positive or negative. Rest days, recent performance, competition type, travel distance to the previous match, teams status (i.e., Elo ranking) are the top contributors to increased match-to-match rotations.

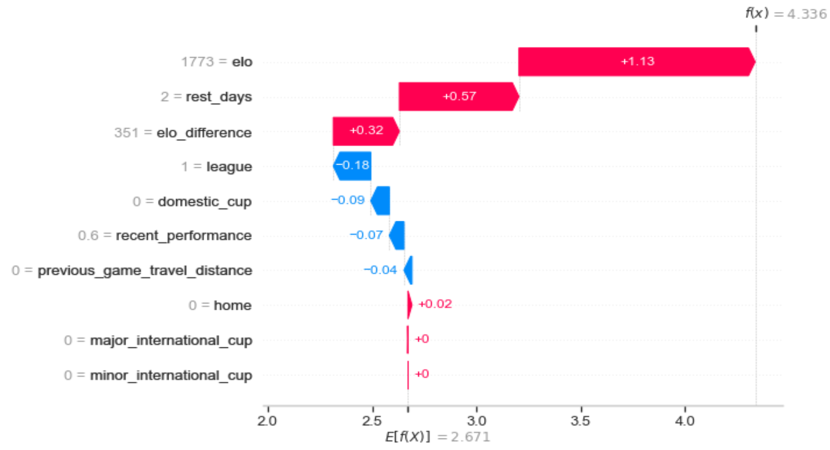

Fig. 5. Contribution of each feature on rotations for Real Madrid vs. Cadiz on May 15, 2022 after playing against Levante on May 12, 2022.

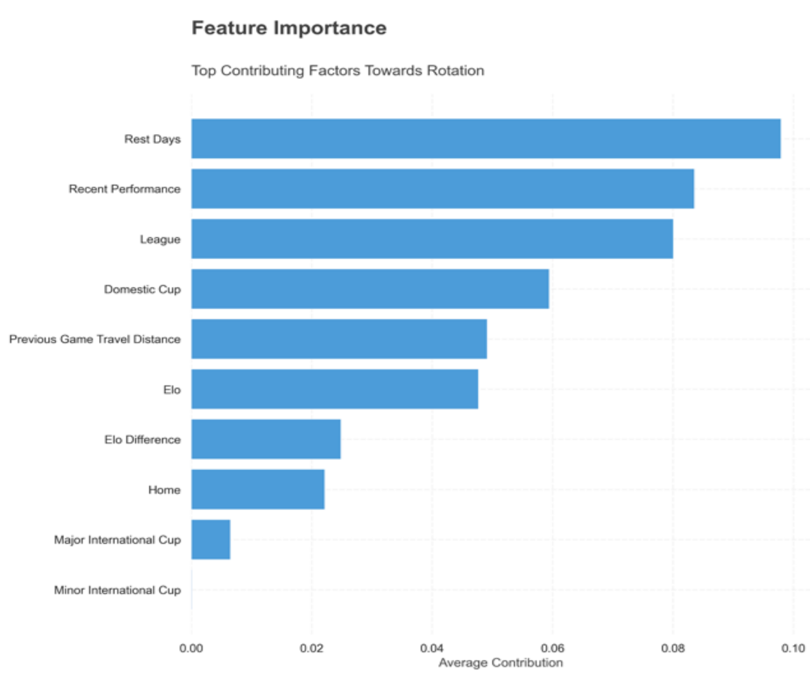

Fig. 6. Feature Importance plot.

Figure 7 provides more details on how the magnitude of these top contributing metrics contributes to match-to-match rotation. In this figure, each data point (game) is shown, and for each metric data point, low values are shaded in blue, and high values in red. If a data point is to the right of the centre line, i.e., a positive SHAP contribution, then it is associated with increased rotation and vice versa. Rest days have a slightly complicated relationship with low and high values leading to increased rotation. More on this below. Recent performance is more straightforward however, and lower, or poorer, recent performance generally leads to increased match-to-match rotation. League is a binary variable, 1 meaning the previous match was a league match, otherwise 0, and there is less match-to-match rotation when the previous match was a league match compared to when it was a domestic cup match.

Figure 8 is a deeper look at rest days, i.e., the number of days between matches, in relation to the contribution towards rotation. It can be seen here that match-to-match rotations were increased when playing the next match within less than three and more than 18 rest days, respectively. This is consistent with the fact that when matches are congested (i.e., <=4-day turnarounds), the coach may rotate more players than usual to avoid increasing their risk of injuries (Page 2023). The increased rotation number following long periods of time (i.e., >10 days) is likely reflective of rotation strategies when playing after an international break. When international players come back at the last minute after extensive travels before their league matches restart, they can't always compete when back.

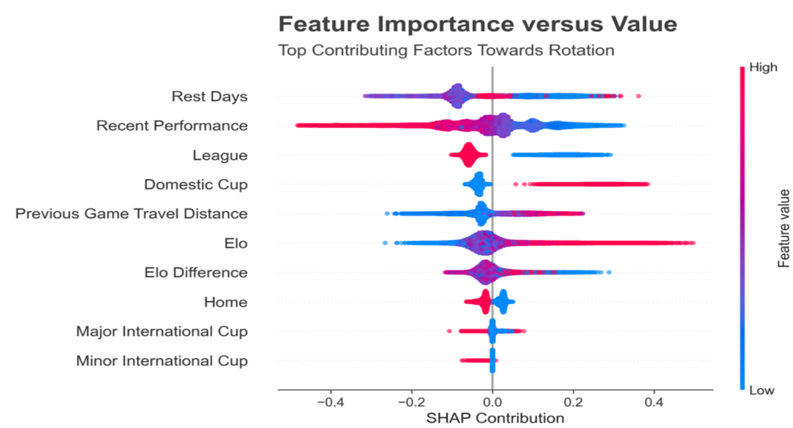

Fig. 7. Feature Relationship plot.

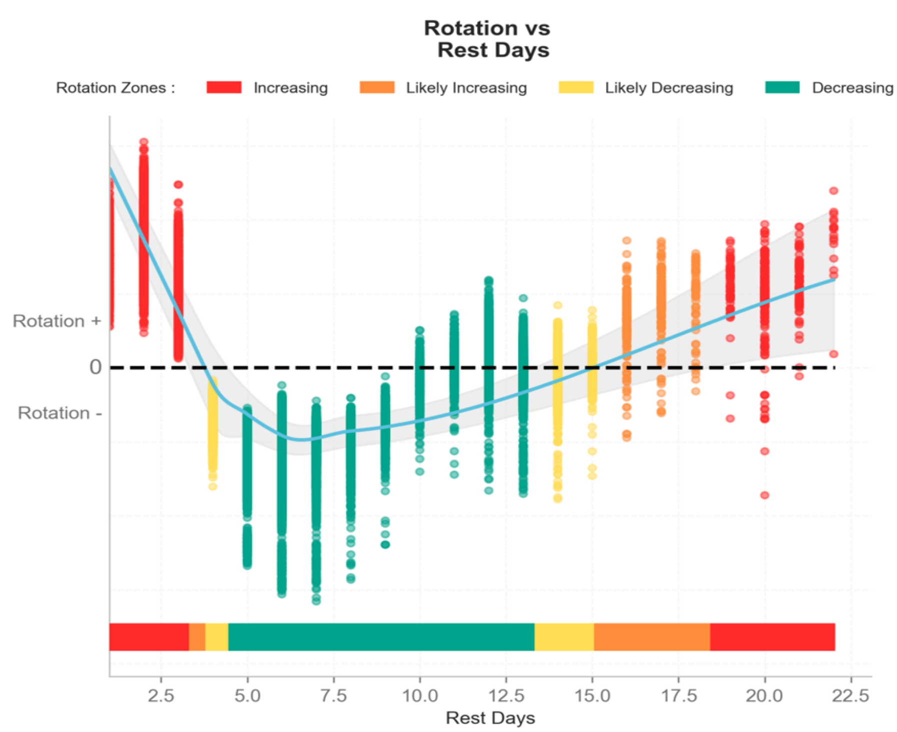

Fig. 8. SHAP dependency plots for rest days (i.e., the number of days separating two consecutive matches).

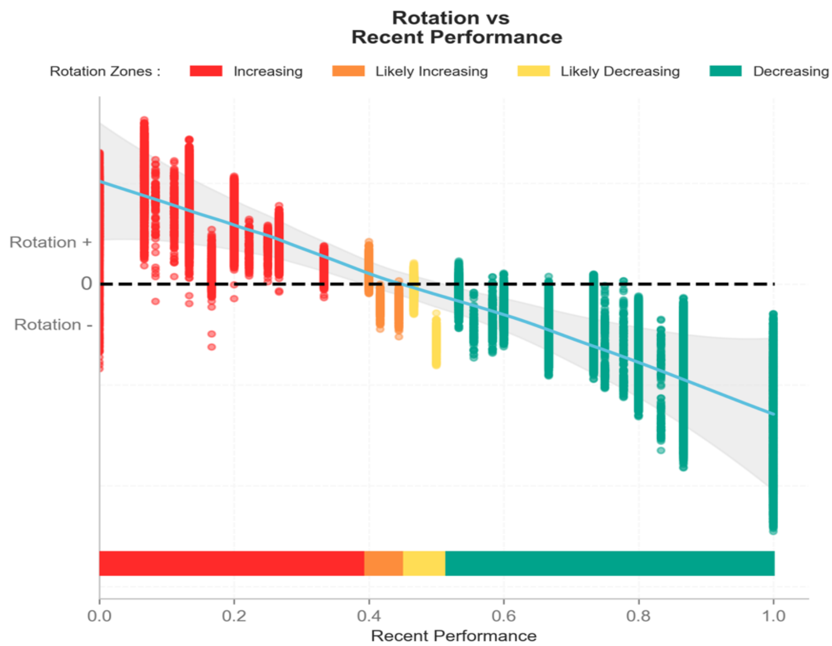

Fig. 9. SHAP dependency plots for recent performance (percentage of possible points won over the last five matches).

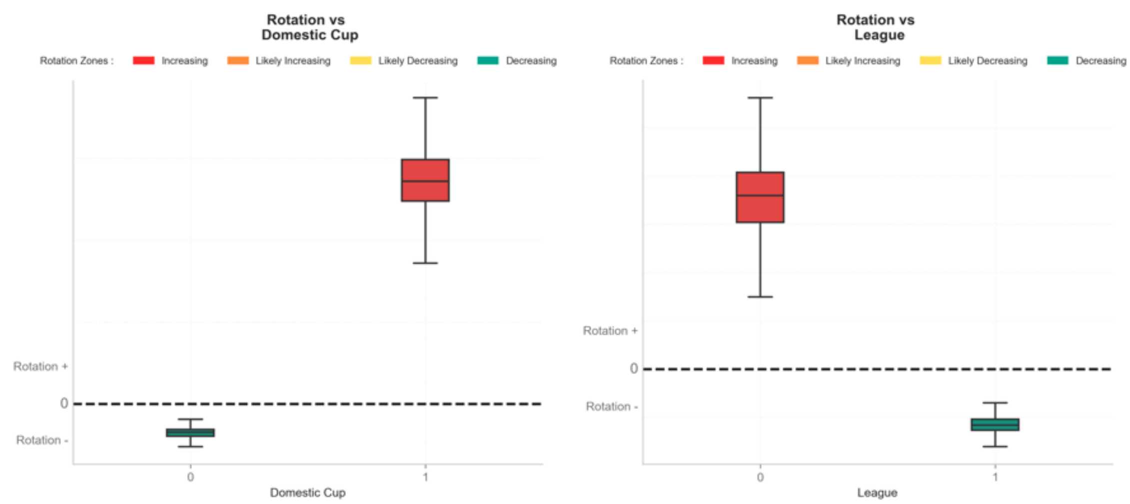

Fig. 10. SHAP dependency plots for last match competition type with a focus on domestic league and cups.

Figure 9 shows recent performance over the last five matches played and also makes intuitive sense, with the worse the recent performance (in terms of percentage of points won), the greater the number of rotations (it's even a linear relationship!). Practically, rotations increased more than normal when a team wins less than 40% of the points available during the last five matches. This is likely related to the fact that when things go well, coaches may not want to change things to keep the dynamic, and conversely, when things don't go too well, changes may help restore performance (10).

It is worth noting that we used a generic definition of "recent performance", which can be interpreted as good or bad for a particular team depending on their context at the time. For example, a team in the relegation zone would be described as being in "good form" if they picked up 10 points from 5 games, but for a title-chasing team the same outcome would probably be considered "bad form". We chose this approach with practicality in mind since quality of performance is team-dependent and may also change with time (and seasons). Defining 'recent performance' at the team level is a topic for future investigation.

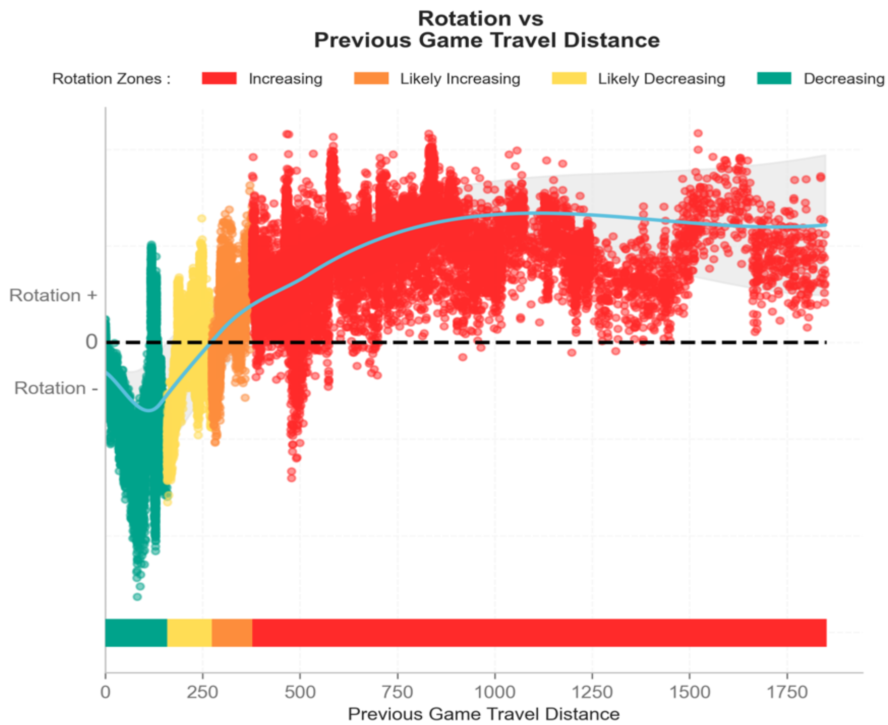

Fig. 11. SHAP dependency plots for previous match travel distance.

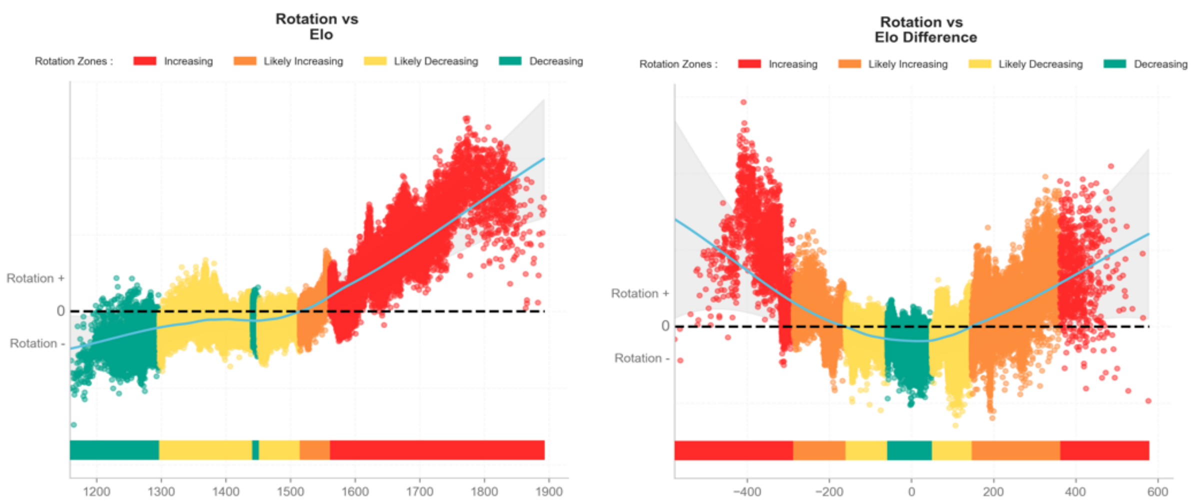

Fig. 12. SHAP dependency plots for the team's Elo (left) and the difference in Elo vs. the opponent (right).

Last match competition type was also a factor influencing rotations (Figure 10), with more and fewer rotations made when playing following a cup match and a league match, respectively. Again, this suggests that while the line-up may be changed transiently for a cup match (which is consistent with common practices, when coaches use those matches as an opportunity to give playing minutes to young players or usual substitutes), line-up stability is favoured when playing the "important" matches such as repeated League matches.

Unsurprisingly as soon as the previous match distance travelled was greater than 370 km, there was an increase in rotations number (Figure 11). If we consider that this distance can be covered by bus in about 3h30-4h maximum, this means that coaches may consider those 4h drives as the cut-off to consider a trip to be detrimental to player freshness.

The contribution of Elo to rotations was straightforward (Figure 12), with top teams (Elo > 1550) rotating more than others, which is likely related to the more extensive depth and higher quality of their squad, which allow them to maintain performance despite changes in players (4). Top teams also generally have more games to play throughout the season, which also increases their need for greater match-to-match rotations. The number of rotations was also decreased when the absolute difference in Elo vs. the opponent was small (<300). These situations likely reflect matches against clubs with similar performance levels/calibre, against which performance is both expected and more likely - coaches would therefore use their best team in this context, i.e., the team with the lowest changes and rotations possible.

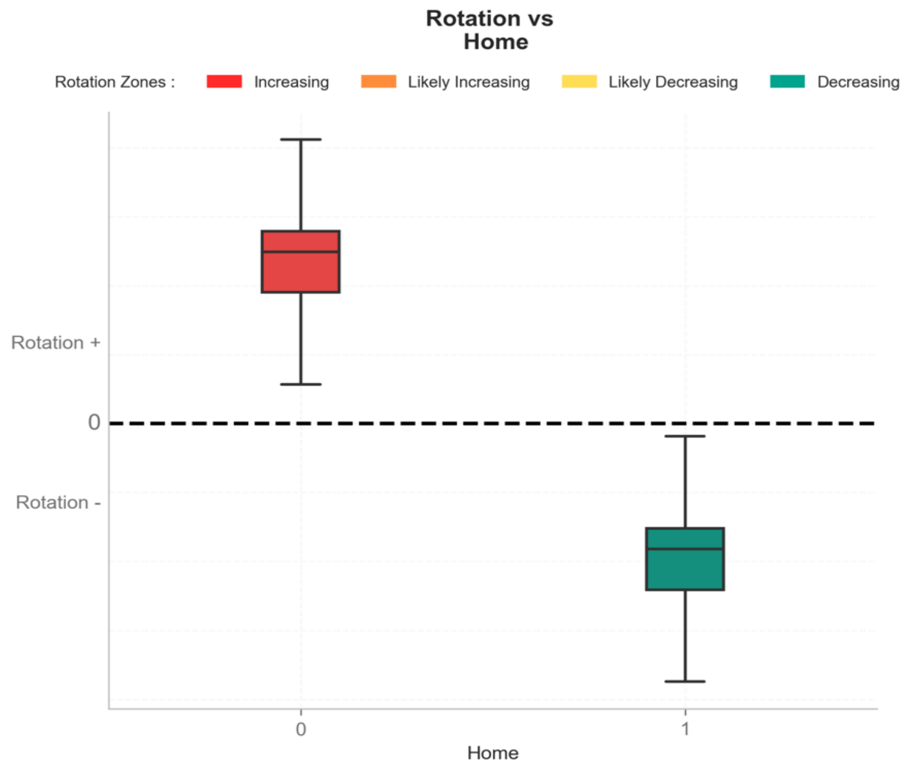

Fig. 13. SHAP dependency plots for match location (home: 1, away: 0).

Rotations were also slightly increased when playing away (Figure 13), irrespective of the previous match location. This may be related to the change in tactics that often occur away, with teams shifting toward more defensive team formations (Ref Part 1). In accordance with the typical home-field advantage (e.g., crowd support, familiar environment, no travel, etc.) (8,11), this increased number of rotations when playing away may partially explain the lower likelihood to win (expected 30% vs. 45% wins, for away vs. home, 4) - considering that rotations potentially induce a decrease in team communication (13).

Overall, rotations are increased at the highest end when top teams (Elo >1550) play away within less than three days following an away (travel distance being >370 km) domestic cup match, when recent performance is poor (<40% of possible points won over the last five matches), and when the absolute difference in Elo vs. their opponent is >300.

The main limitation of the current analysis remains the data set obtained from an online database. This data set constrained the factors included in the modelling part, since there may be other factors that were not available from Transfermarkt, whose influence could therefore not be examined.

Key findings

- There is a tradeoff between line-up stability for optimal match performance and load management to preserve players' health when it comes to deciding on how many players to rotate from one match to the next, and who to rotate.
- Overall match-to-match rotation numbers were between two and three per team and not clearly different between leagues, with coaches of the Ligue 1/Serie A and the Eredivisie tending to perform the most (around three) and the least (less than two) number of rotations, respectively.
- There were some small increasing trends in match-to-match rotations across the years in some leagues only (i.e., from two to three in the Premier League and La Liga) but no apparent differences between home and away matches.
- The key factors that could affect coaches' rotation strategies were ranked in relation to their respective contribution to the model - with rest days, recent performance, competition type, travel distance to the previous match, teams status (i.e., Elo ranking) being the top contributors to increased match-to-match rotations.
- To summarise, rotations are increased at the highest end when top teams (Elo >1550) play:
  1. With fewer than three rest days
  2. In the context of poor recent performances (<40% of possible points won over the last 5 matches)
  3. Following an away (travel distance being >370 km) domestic cup match
  4. Against an opponent considerably worse or better than them (absolute difference in Elo vs. their opponent >300)
  5. Away

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Appendix A

Elo Rating System

The Elo Rating System is a method for calculating the relative strength-levels of competitors in zero-sum games, and was originally used to rank players in chess. Our football implementation uses data from transfermarkt.com to rate teams in more than 40 countries worldwide, based on all available competitive matches from as early as 1980. Each team's rating is updated dynamically after each game, so at any point in time a team's rating represents their strength-level relative to all other teams within the system.

Before a game, the difference in the ratings between the two teams, along with a home-field advantage adjustment, is used to estimate the probability of the match ending in a home win, away win, or draw. After the game, the estimated outcome probabilities are compared to the observed outcomes and Elo Rating Points are taken from the losing team and awarded to the winning team in proportion to how likely they were expected to win. In this way the ratings are self-correcting, as an underrated team will achieve better outcomes than expected and therefore earn a higher rating that more accurately reflects their ability.

The amount of Elo Rating Points available to be exchanged post-game is controlled by a parameter known as the K-Factor, and we adjust this dynamically to take into account some additional factors. The K-Factor is raised (meaning more rating points are exchanged) in continental cup games (e.g., Champions League) and lowered in domestic cup matches (e.g., EFL Cup). We also increase the K-Factor in games earlier in the dataset to allow the team ratings to update more quickly from their initial starting points.